

Statistical and Computational Models

Lecture 3: Nonparametric regression

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Roadmap

Beyond a fixed form

One tuning parameter

Training and test MSE

Parametric vs. Non-Parametric Models

PARAMETRIC

Assume a form: $f(\mathbf{x}) = \beta_0 + \beta_1 \mathbf{x}$.

Estimate $p + 1$ parameters.

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Non-parametric models can reduce approximation error, but like in life there is **no free lunch...**

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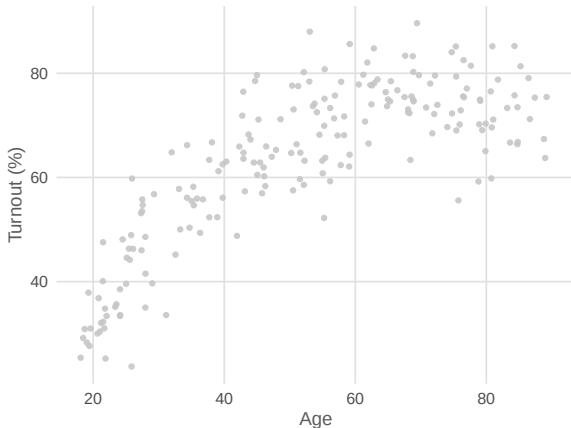
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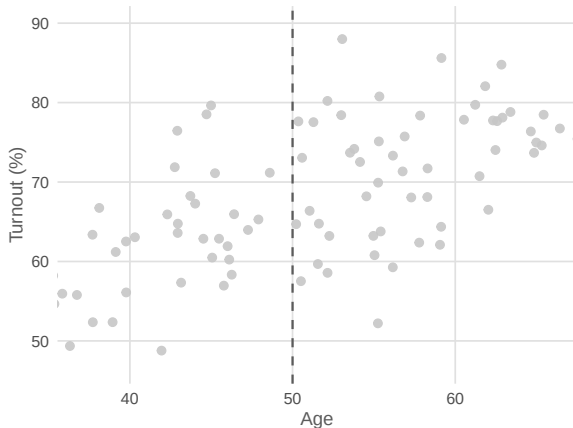
There is no β to estimate, but we have to pick k . How?

Building the k-nn regression line



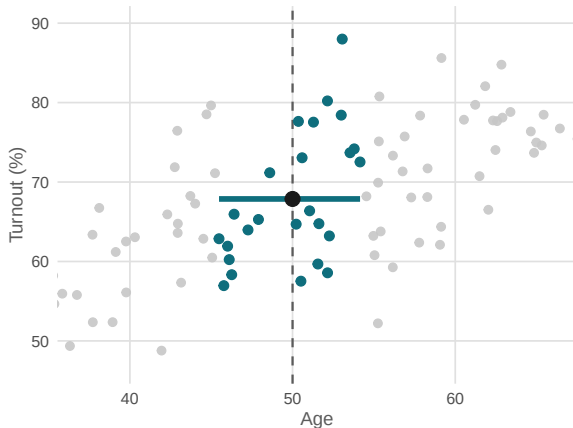
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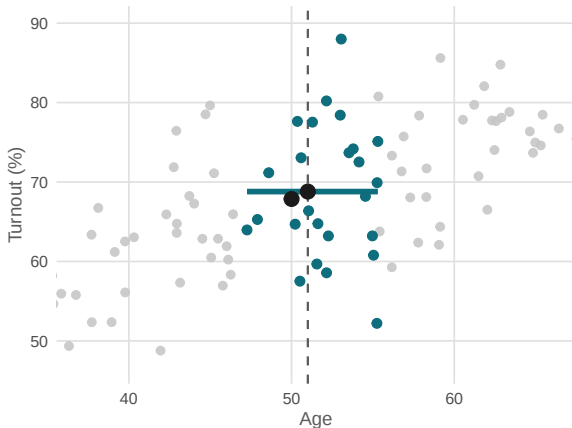
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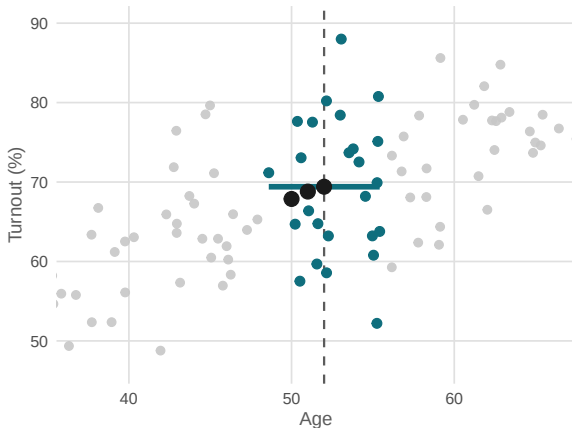
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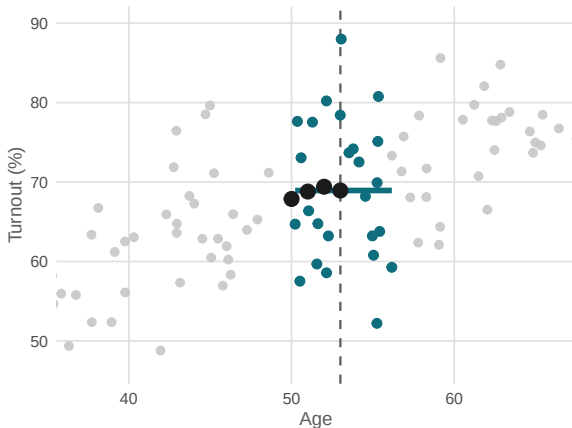
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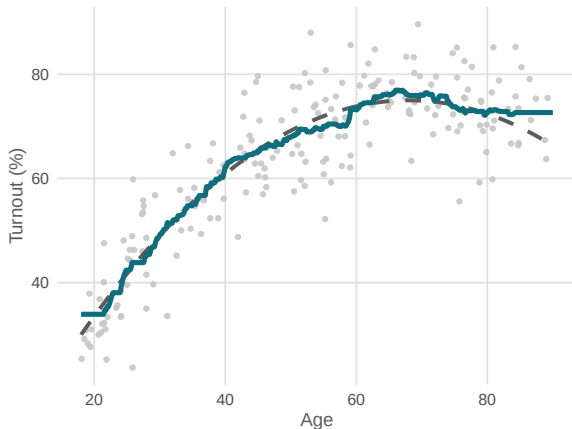
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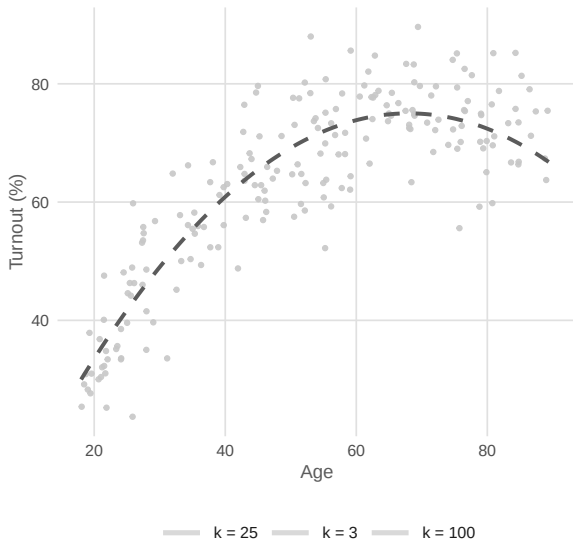
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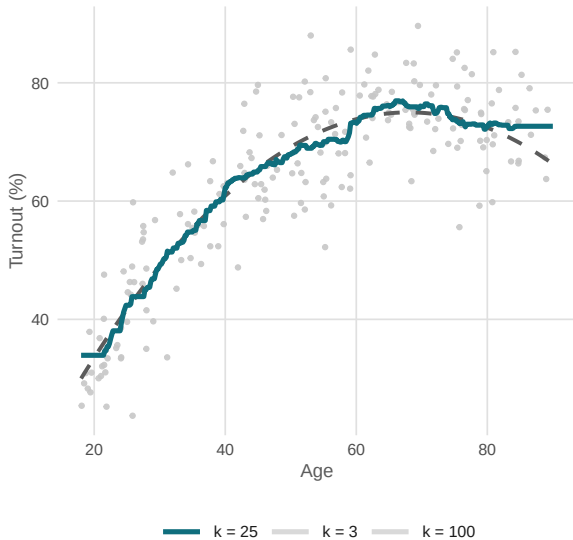
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Do that at every age and you get back a line. But **why $k = 25$?**

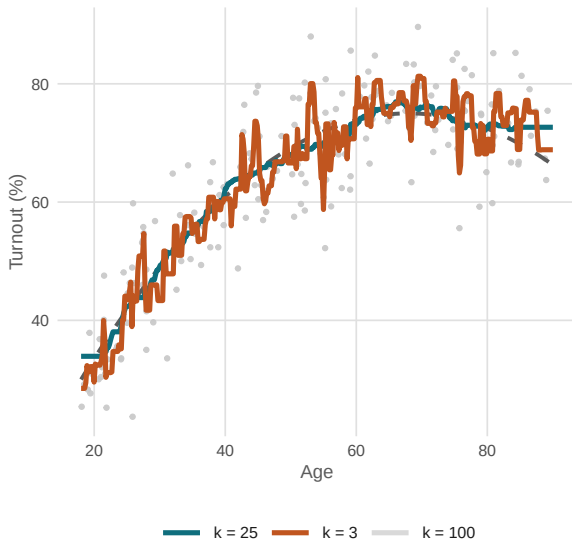
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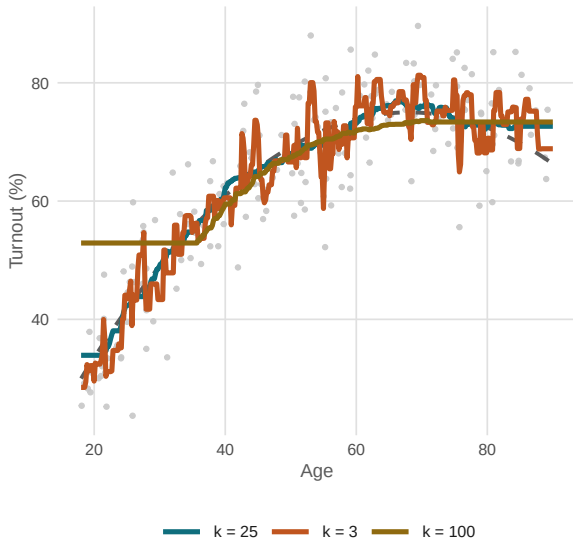
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Uncertainty will not have a closed-form solution, so **resample**, e.g. a bootstrap.

Roadmap

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One tuning parameter

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What k controls

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LARGE k

Many neighbours.

We smooth between ages.

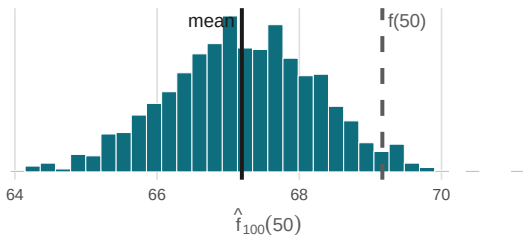
High **bias**.

Bias and variance

Suppose we have a proposed estimator $\hat{f}(x_0)$. Draw a sample, compute $\hat{f}(x_0)$. Draw another, compute it again. Over repeated samples $\hat{f}(x_0)$ has a **sampling distribution**.

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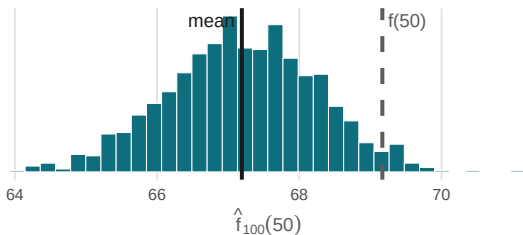
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$$\text{Bias}(x_0) = \mathbb{E}[\hat{f}(x_0)] - f(x_0), \quad \text{Var}(x_0) = \mathbb{E}\left[\left(\hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2\right]$$

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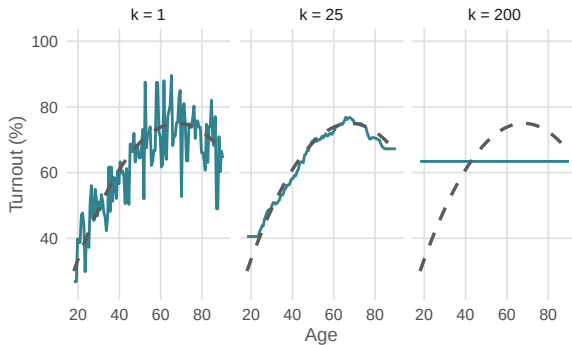
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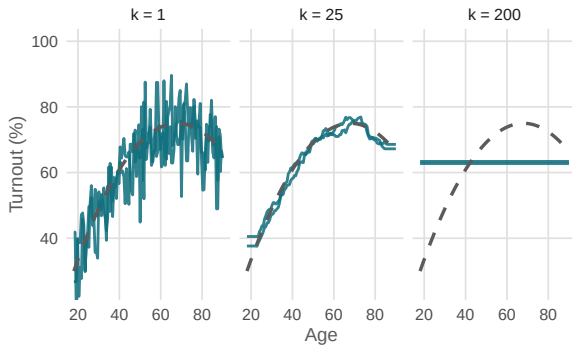
Read: bias is the **central tendency** of that distribution, against the estimand. Variance is its **spread**.

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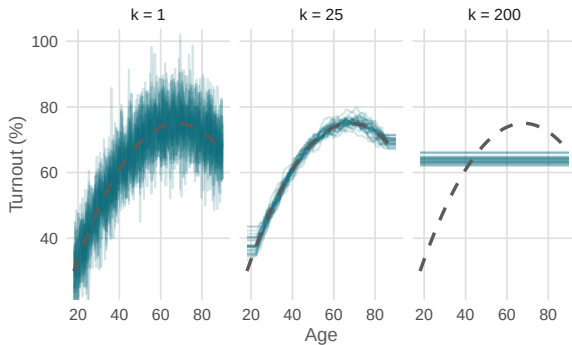
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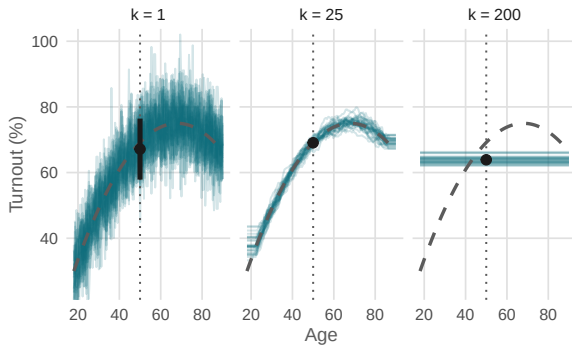
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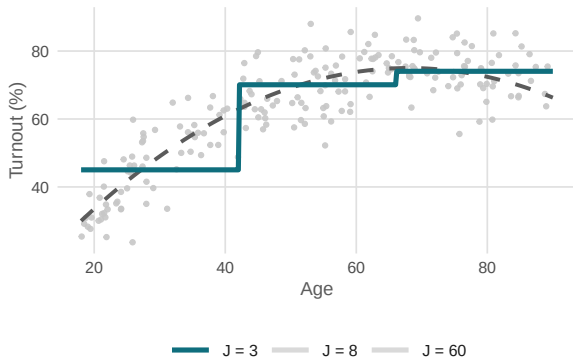
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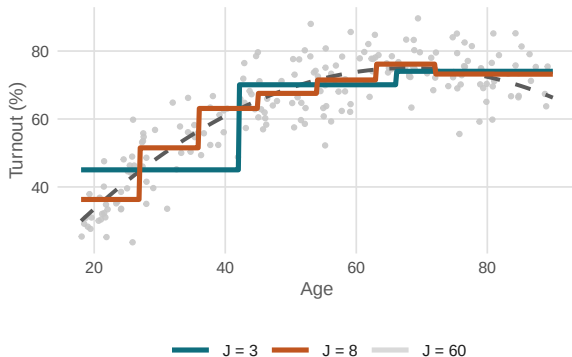
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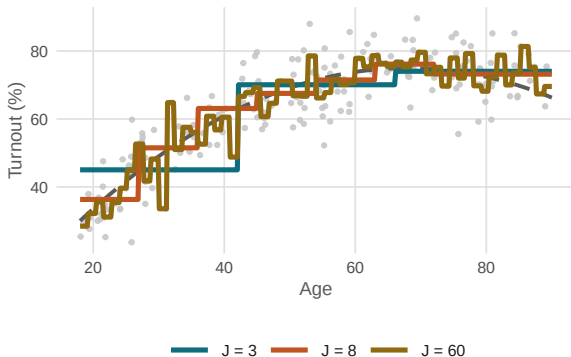
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Move the bin with $\mathbf{x}_0$ and it is a **local smoother**. Weight by distance or a **kernel** (we will see this in a later lecture).

Back to the linear model

One more detour. What if we propose a linear regression but make its **order** the tuning parameter.

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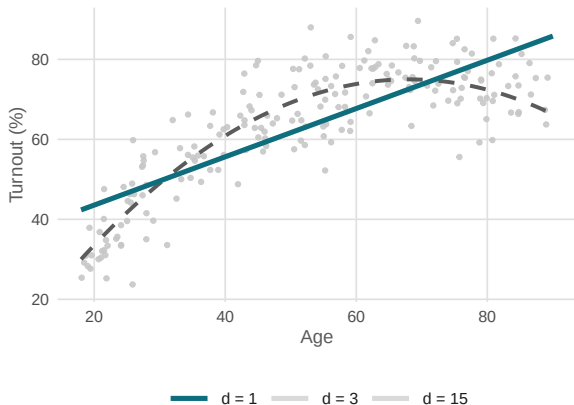
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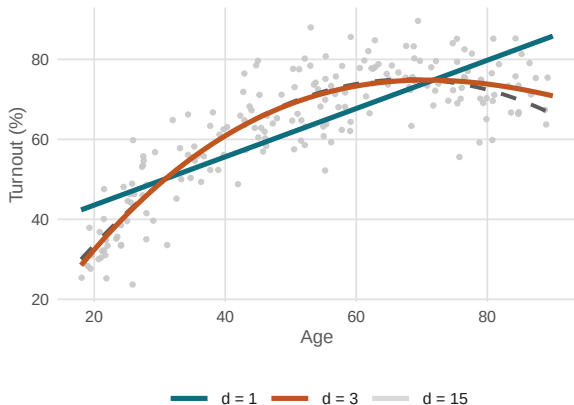
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$d = 1$ is a straight line, $d = 3$ the cubic form, and so on.

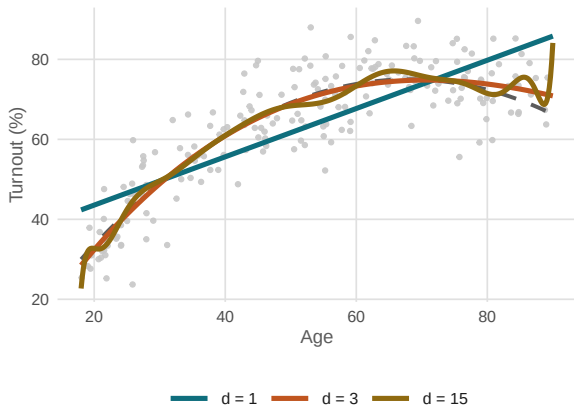
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Is the **wiggliest one the best**? It turns out this is the same problem as picking k , or picking J .

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Training vs. test vs. EPE

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Training vs. test vs. EPE

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We cannot see the EPE, so we need an empirical alternative:

Quantity	Averages the loss over	Observable?
Training loss $\frac{1}{n} \sum_{i=1}^n L(y_i, \hat{f}(x_i))$	the n observations we fit $\hat{f}$ on	always
Test loss $\frac{1}{m} \sum_{j=1}^m L(y_j, \hat{f}(x_j))$	m observations not used in fitting $\hat{f}$	only if you have unseen data
EPE $\mathbb{E}[L(Y, \hat{f}(X))]$	the population (or very many draws from the DGP)	never

Two errors

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Always!

Why training MSE lies

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$$\mathbb{E}[\text{MSE}_{\text{train}}(\hat{f})] \leq \text{EPE}(f^\dagger) \leq \mathbb{E}[\text{EPE}(\hat{f})]$$

QED

Why training MSE lies

Let $f^\dagger$ minimise EPE over the class, and $\hat{f}$ minimise $\text{MSE}_{\text{train}}$.

$\hat{f}$ was chosen to minimise $\text{MSE}_{\text{train}}$ for the realised sample. So $\text{MSE}_{\text{train}}(\hat{f}) \leq \text{MSE}_{\text{train}}(f^\dagger)$.

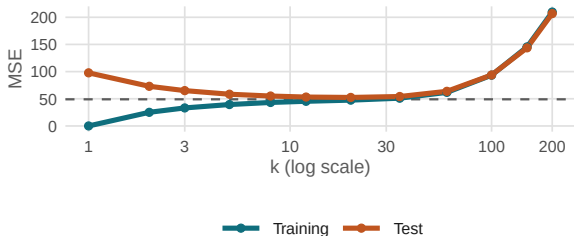
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QED

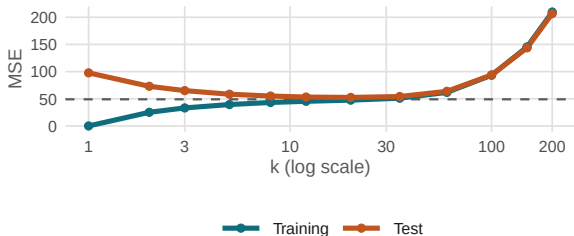
At $k = 1$ the fit runs through every observation: $\text{MSE}_{\text{train}} = 0$ (a perfect model!).

The U-curve



Let's see this in action. Over 40 samples, calculate both the test and the train curves.

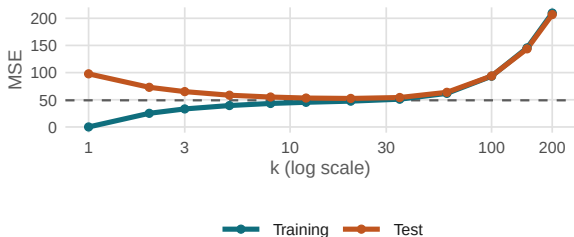
The U-curve



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$\hat{f}$ **overfits** when some less flexible fit would give lower test MSE.

The U-curve



Let's see this in action. Over 40 samples, calculate both the test and the train curves.

$\hat{f}$ **overfits** when some less flexible fit would give lower test MSE.

So, how do we pick k or J or d ? We'll learn that next week.

References

- James, G., Witten, D., Hastie, T. & Tibshirani, R. (2021) *An Introduction to Statistical Learning*, 2nd ed. – **3.5, 7.1–7.2**. statlearning.com.
- Hastie, T., Tibshirani, R. & Friedman, J. (2009) *The Elements of Statistical Learning*, 2nd ed. – 2.3, 7.1–7.3.